KLH BACHUPALLY

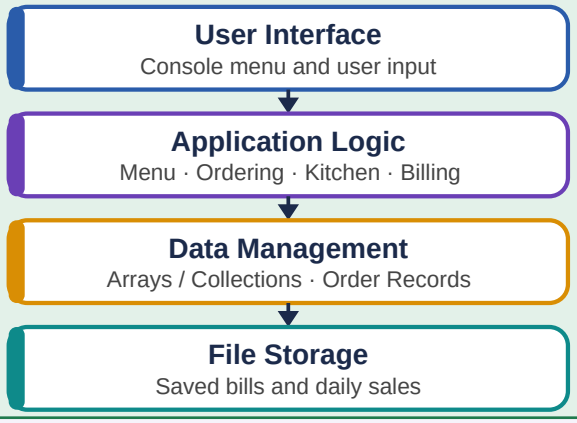
Restaurant Order, Kitchen and Billing System

PROJECT BASED LEARNING

Abstract

The **Restaurant Order, Kitchen and Billing System** is a menu-driven **Java** application designed to simplify restaurant operations. It enables users to view menu items, place orders, manage quantities, and generate bills automatically. The system aims to reduce manual effort and billing errors while improving operational efficiency. The project demonstrates Java concepts such as variables, control statements, methods, arrays, classes, objects, exception handling, and data management. Future enhancements may include kitchen status tracking, persistent transaction records, and advanced sales analysis.

System Architecture



CO-WISE IMPLEMENTATION

CO1 Java Foundations

Data types, variables, expressions and console I/O

Stores food names, prices, quantities, table numbers and bill amounts. Accepts user input and calculates order totals.

CO2 Control Flow

Selection, switch-case and loops

Uses a menu-driven interface, validates user choices, and repeats operations until the user exits.

CO3 Methods, Recursion and Arrays

Methods and arrays

Uses separate methods for displaying menus, taking orders and generating bills. Arrays store item names, prices and quantities.

CO4 OOP and Exception Handling

Classes, objects, encapsulation and exceptions

Models menu items and orders using classes and objects. Handles invalid input and other runtime errors using exception handling.

CO5 Strings, Algorithms and File I/O

String methods, searching, sorting and files

Processes food names, searches menu items and saves order records or daily sales to a file.

CO6 Collections and Functional Programming

Collections, generics, lambdas and streams

Uses collections to manage orders and streams or lambda expressions to filter orders and summarize sales.

Conclusion & Future Scope

The project demonstrates the practical application of Java programming concepts in automating basic restaurant operations. It provides a structured approach to order management, kitchen processing, and billing while improving accuracy and reducing manual effort.

Future enhancements may include a graphical user interface, database integration, online ordering, digital payments, inventory tracking, and real-time kitchen status updates.

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